

I-Chun (Arthur) Liu

San Francisco Bay Area, CA, United States

✉ arthurleyu@gmail.com | 🏠 arthurliu.com | 📞 arthur801031 | 📺 i-chun-arthur-liu | 🎓 Google Scholar

Education

University of Southern California

DOCTOR OF PHILOSOPHY, COMPUTER SCIENCE (ROBOTICS & AI)

Aug 2023 - Dec 2026 (Expected)

Los Angeles, CA

University of Southern California

MASTER OF SCIENCE, COMPUTER SCIENCE

Jan 2021 - May 2023

Los Angeles, CA

University of California, Los Angeles

MASTER OF SCIENCE, COMPUTER SCIENCE

Oct 2014 - Jun 2016

Los Angeles, CA

University of Wisconsin-Madison

BACHELOR OF SCIENCE, COMPUTER SCIENCE

Sep 2010 - May 2014

Madison, WI

Publications

- [1] **I-Chun Arthur Liu***, Jason Chen*, Gaurav S. Sukhatme, and Daniel Seita. ExStereo: Lifting 2D Vision-Language-Action Models to 3D with Explicit Stereo Representations. In *Submission*, 2026.
- [2] **I-Chun Arthur Liu**, Krzysztof Choromanski, Sandy Huang, and Connor Schenck. CLAMP: Contrastive Learning for 3D Multi-View Action-Conditioned Robotic Manipulation Pretraining. In *Robotics: Science and Systems (RSS)*, 2026.
- [3] Jason Chen, **I-Chun Arthur Liu**, Gaurav S. Sukhatme, and Daniel Seita. CRAFT: Video Diffusion for Bimanual Robot Data Generation. In *International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [4] Jason Chen, **I-Chun Arthur Liu**, Gaurav S. Sukhatme, and Daniel Seita. ROPA: Synthetic Robot Pose Generation for RGB-D Bimanual Data Augmentation. In *International Conference on Robotics and Automation (ICRA)*, 2026.
- [5] **I-Chun Arthur Liu**, Jason Chen, Gaurav S. Sukhatme, and Daniel Seita. D-CODA: Diffusion for Coordinated Dual-Arm Data Augmentation. In *Conference on Robot Learning (CoRL)*, 2025.
- [6] **I-Chun Arthur Liu**, Sicheng He, Daniel Seita[†], and Gaurav S. Sukhatme[†]. VoxAct-B: Voxel-Based Acting and Stabilizing Policy for Bimanual Manipulation. In *Conference on Robot Learning (CoRL)*, 2024.
- [7] Gautam Salhotra*, **I-Chun Arthur Liu***, and Gaurav S. Sukhatme. Learning Robot Manipulation from Cross-Morphology Demonstration. In *Conference on Robot Learning (CoRL)*, 2023.
- [8] Gautam Salhotra*, **I-Chun Arthur Liu***, Marcus Dominguez-Kuhne, and Gaurav S. Sukhatme. Learning Deformable Object Manipulation from Expert Demonstrations. In *IEEE Robotics and Automation Letters (RA-L)* and *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [9] **I-Chun Arthur Liu***, Shagun Uppal*, Gaurav S. Sukhatme, Joseph J. Lim, Peter Englert, and Youngwoon Lee. Distilling Motion Planner Augmented Policies into Visual Control Policies for Robot Manipulation. In *Conference on Robot Learning (CoRL)*, 2021.

* denotes equal contribution. [†] denotes equal advising.

Research Experience

Google DeepMind

STUDENT RESEARCHER. HOSTED BY CONNOR SCHENCK.

- VLA research.

June 2026 - Present

Mountain View, CA, United States

Google DeepMind

STUDENT RESEARCHER. HOSTED BY CONNOR SCHENCK.

- 3D pre-training framework for robotic manipulation (RSS 2026).

June 2025 - Dec 2025

Mountain View, CA, United States

Amazon Robotics

APPLIED SCIENTIST INTERN

- Targeted bin picking research (robotic manipulation).

May 2023 - August 2023

MA, United States

University of Southern California, Robotics Embedded Systems Laboratory (RESL)

January 2021 - May 2023

RESEARCH ASSISTANT (PART-TIME, PAID POSITION). ADVISED BY PROF. GAURAV SUKHATME.

Los Angeles, CA, United States

- Published three papers (CoRL 2021, RA-L + IROS 2022, CoRL 2023) as first author. Research interests: reinforcement learning and vision for robotics.

Academia Sinica (Institute of Information Science), Multimedia Technologies Laboratory

January 2019 - December 2020

RESEARCH ASSISTANT (FULL-TIME, PAID POSITION). ADVISED BY PROF. HONG-YUAN MARK LIAO AND PROF. LI-FEN CHEN.

Taipei, Taiwan

- Conducted research on computer vision in topics including object detection and instance segmentation.

Achievements

2025 - USC URAP Funding, Assisted my advisee, Jason, in securing this funding to support his work on my research project.

2023 - USC 1-Year Fellowship, University of Southern California - Viterbi School of Engineering (awarded to selected PhD students).

2023 - Computer Science Best Research Award, University of Southern California - Viterbi School of Engineering (MS Award).

Mentoring

USC Undergraduate Students: Jason Chen (CoRL 2025, ICRA 2026, IROS 2026, Current)

USC Masters Students: Sicheng He (CoRL 2024) → PhD at UC Irvine, Hanwen Fan (Current) → PhD at Iowa State University

External Collaborators: QJ Tsway (Current)

Talks

2026 ICLR Expo Talk with the Robot Vision Team from Google DeepMind: invited talk on CLAMP (Apr.)

Google DeepMind: research presentation on CLAMP (Mar.)

USC PhD Thesis Proposal: “3D Representations for Sample-Efficient Robotic Bimanual Manipulation” (Feb.)

2025 Google DeepMind: research presentation on D-CODA (Jul.)

USC PhD Qualifying Exam: “VoxAct-B: Voxel-Based Acting and Stabilizing Policy for Bimanual Manipulation” (Mar.)

Services

Reviewer CoRL 2022 to 2025; WAFR 2024; ICRA 2025 to 2026; IROS 2026; NeurIPS 2026

Teaching

Spring 2025, Teaching Assistant - USC CSCI 572: Information Retrieval and Web Search Engines (Master's Level), Saty Raghavachary.

Fall 2024, Teaching Assistant - USC CSCI 545: Introduction to Robotics (Master's Level), Stefanos Nikolaidis.

Work Experience

Appier

October 2017 - April 2018

SOFTWARE ENGINEER, FRONTEND

Taipei, Taiwan

- Maintained and developed new features for dynamic creatives generator tool using React, Redux, JavaScript, and CSS.

Mozilla (Firefox DevTools Debugger)

November 2016 - November 2017

OPEN SOURCE CONTRIBUTOR

GitHub

- Made 35 contributions, from small UI fixes to significant features, to Firefox's new DevTools Debugger, which is written in React and Redux.

National Development Council

August 2016 - August 2017

FOREIGN LANGUAGE SPECIALIST — CONSCRIPT

Taipei, Taiwan

- Translated social/cultural development policy documents from and into Chinese and English.